

Lectures On Class Field Theory

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1 Un peu d'histoire

We start with some motivation: this course will be about number fields (i.e. finite extensions $K|\mathbf{Q}$) and their absolute Galois groups $G_K = \text{Gal}(\overline{K}|K)$, where $\overline{K}$ is a fixed algebraic closure of K . Very little is known about the structure of such automorphism groups in general, and a large chunk of modern number theory/arithmetic geometry is devoted to studying them, e.g. via their representations. In some sense, class field theory consists in the study of its 1-dimensional representations; more precisely, it describes the abelianization $G_K^{\text{ab}} = G_K/[G_K, G_K]$ of the absolute Galois group (equivalently, of its abelian quotients).

1.1 Reciprocity laws

Let p be a prime number, and let a be an integer coprime to p . The *Legendre symbol* $\left(\frac{a}{p}\right)$ is defined as

$$\left(\frac{a}{p}\right) = \begin{cases} 1 & \text{if } a \text{ is a square modulo } p, \\ -1 & \text{if } a \text{ is not a square modulo } p. \end{cases}$$

Euler proved that $\left(\frac{a}{p}\right) = a^{\frac{p-1}{2}} \pmod{p}$, and **Gauss** proved the following law:

Theorem 1.1 (Quadratic reciprocity). Let $a \neq b$ be different odd primes. Then,

$$\left(\frac{a}{b}\right) \cdot \left(\frac{b}{a}\right) = (-1)^{\frac{a-1}{2} \frac{b-1}{2}}.$$

Moreover, the following subsidiary laws hold:

1. $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}};$
2. $\left(\frac{2}{p}\right) = (-1)^{\frac{p^2-1}{8}}.$

These laws, together with its multiplicativity in the parameter a , make it possible to compute the Legendre symbol.

Another way of thinking about class field theory is as an attempt to define a “very general” reciprocity law, from which the quadratic reciprocity law and its higher degree analogues can be deduced. One such generalization is **Hilbert’s** reciprocity law: recall that $\mathbf{Q}$ embeds into the p -adic field $\mathbf{Q}_p$ for every prime p , and also into the real numbers $\mathbf{Q}_\infty := \mathbf{R}$. For every prime p (including $p = \infty$) and for all $a, b \in \mathbf{Q}^\times$, we define the *Hilbert symbol* $(a, b)_p$ as

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ has a solution in } \mathbf{Q}_p^2, \\ -1 & \text{otherwise.} \end{cases}$$

Recall that the norm $N : \mathbf{Q}_p(\sqrt{a}) \rightarrow \mathbf{Q}_p$ is defined as $N(x + y\sqrt{a}) = x^2 - ay^2$ if $a \notin \mathbf{Q}_p^{\times 2}$, and as $N = \text{id}$ otherwise. More generally, if $L|K$ is a finite Galois extension with group G , the norm $N_{L|K} : L \rightarrow K$ is

$$N(x) = \prod_{\sigma \in G} \sigma(x)$$

Lemma 1.2. The following statements are equivalent:

1. $(a, b)_p = 1$;
2. The quadratic equation $z^2 = ax^2 + by^2$ has a non-trivial solution;
3. $a \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{b})$.
4. $b \in \mathbf{Q}_p$ is the norm of an element of $\mathbf{Q}_p(\sqrt{a})$.

Proof. The equivalence $1 \iff 2$ is by definition; and $3 \iff 4$ is clear. For $2 \iff 3$, if $b \in \mathbf{Q}_p^{\times 2}$, then we are done. Otherwise, if $b \notin \mathbf{Q}_p^{\times 2}$, then $N(\alpha + \beta\sqrt{b}) = \alpha^2 - b\beta^2$, and if $a := \alpha^2 - b\beta^2$ we see that $z^2 = ax^2 + by^2$ has a non-trivial solution, and vice versa. $\square$

For $p \neq 2, \infty$ and $a, b \in \mathbf{Z}$, write $a = p^{v_p(a)}u$ and $b = p^{v_p(b)}v$ with $p \nmid uv$. The relation between the Hilbert symbol and the Legendre symbol is

$$(a, b)_p = (-1)^{\frac{p-1}{2}v_p(a)v_p(b)} \left(\frac{u}{p}\right)^{v_p(b)} \left(\frac{v}{p}\right)^{v_p(a)} \quad (1.1)$$

$$= \overline{(-1)^{v_p(a)v_p(b)} a^{v_p(b)} b^{-v_p(a)}}^{(p-1)/2}. \quad (1.2)$$

where the bar denotes reduction modulo p . In particular, if p does not divide a , then $(a, b)_p = \left(\frac{a}{p}\right)^{v_p(b)}$ coincides with the Legendre symbol.

Theorem 1.3 (Hilbert reciprocity law). For all $a, b \in \mathbf{Q}^\times$, we have

$$\prod_p (a, b)_p = 1$$

As for computation, one shows that if p does not divide $2ab$ (including $p = \infty$ for $a, b > 0$), then $(a, b)_p = 1$. Moreover, $(a, b)_2 = (-1)^{\frac{a-1}{2}\frac{b-1}{2}}$ for odd a, b . From this and multiplicativity in the first variable, we immediately deduce the quadratic reciprocity law by taking a, b to be odd primes.

An advantage of this formulation is that it generalizes to number fields K , by considering the completions of K at all of its (finite and infinite) “places”. Hilbert then went on to pose his 9th problem, in which he asked for a generalization of this law to n -th powers in arbitrary number fields; this is what class field theory achieves. We first consider the case of extensions which contain a primitive n -th root of unity:

1.2 Kummer case

Assume now $K|\mathbf{Q}$ is a finite extension such that $\mu_n \subset K$, and let $\mathcal{O}_K$ be its ring of integers. If $P \subset \mathcal{O}_K$ is a prime ideal, we would like to define a symbol $(a, b)_P \in \mu_n$ such that a product formula holds, i.e.

$$\prod_P (a, b)_P = 1$$

if $a, b \in K^\times$. Moreover, for P finite (which is always the case for $n \neq 2$) which does not divide (n) we shall have

$$(a, b)_P = \overline{(-1)^{v_P(a)v_P(b)} a^{v_P(b)} b^{-v_P(a)} (q-1)^{1/n}},$$

where $q = \#(\mathcal{O}_K/P)$. In particular, if P does not divide (a) , then $(a, b)_P = a^{\frac{q-1}{n} v_P(b)} \bmod P$.

So, if $v_P(b) = 1$, then

$$\left(\frac{a}{P}\right) := (a, b)_P$$

does not depend on the choice of b and it is the unique n -th root of unity, that is $a^{\frac{q-1}{n}} \bmod P$; in other words, $\left(\frac{a}{P}\right) = 1$ if and only if $x^n = a \bmod P$ has a solution. Some more properties of this symbol are:

1. $(a, b)_P = 1 \iff b$ is a norm from $K_P(\sqrt[n]{a}) \iff a$ is a norm from $K_P(\sqrt[n]{b})$;
2. $(a, b)_P$ is multiplicative in both variables and anticommutative;

When μ_n is *not* contained in K_P , an “Hilbert” symbol still exists but is longer associated to $a, b \in K^\times$. Suppose for a moment that $\mu_n \subset K_P$. Then, the extension $K_P(\sqrt[n]{a})|K_P$ is Galois for all $a \notin K_P^{\times n}$, and its Galois group is

$$G_a := \text{Gal}(K_P(\sqrt[n]{a})|K_P) \cong \mathbf{Z}/n\mathbf{Z}$$

Then, the group $\text{Hom}(G_a, \mu_n) \cong \mathbf{Z}/n\mathbf{Z}$ is generated by the character

$$\overline{\chi}_a : \sigma \mapsto \frac{\sigma(\sqrt[n]{a})}{\sqrt[n]{a}} \in \mu_n \cong \mathbf{Z}/n\mathbf{Z}.$$

In general, we shall construct a generalized Hilbert symbol

$$\begin{aligned} \text{Hom}(\text{Gal}(\overline{K_P}|K_P), \mathbf{Z}/n\mathbf{Z}) \times K_P^\times &\rightarrow \mathbf{Z}/n\mathbf{Z}, \\ [\chi, b] &\mapsto (\chi, b)_P \end{aligned}$$

and we'll thus be able to prove the **Artin** reciprocity law: fix some $b \in K^\times$, and suppose that χ is a character in $\text{Hom}(G_K, \mathbf{Z}/n\mathbf{Z})$. Then

$$\sum_P (\chi_P, b) = 0,$$

where χ_P is the restriction of χ to $G_{K_P}^{\text{ab}}$ (recall that it embeds into G_K^{ab}).

The generalized Hilbert symbol defines a reciprocity map

$$K_P^\times \rightarrow G_{K_P}^{\text{ab}},$$

which is “almost” an isomorphism. In particular, for every n we'll have an isomorphism of finite groups

$$K_P^\times/n \rightarrow G_{K_P}^{\text{ab}}/n.$$

This is one of the remarkable features of class field theory, giving a description of the abelian extensions of a field in terms of its arithmetic. To define such a map, it's enough to define the dual map

$$\begin{aligned} \text{Hom}(G_{K_P}^{\text{ab}}/n, \mathbf{Z}/n\mathbf{Z}) &\rightarrow \text{Hom}(K_P^\times/n, \mathbf{Z}/n\mathbf{Z}), \\ [\chi] &\mapsto [b \mapsto (\chi, b)_P]. \end{aligned}$$

2 Computing the Hilbert symbol

Recall the definition: for $a, b \in \mathbf{Q}_p^\times$,

$$(a, b)_p = \begin{cases} 1 & \text{if } ax^2 + by^2 = 1 \text{ is solvable in } \mathbf{Q}_p, \\ -1 & \text{otherwise.} \end{cases}$$

We have seen that

$(a, b)_p = 1 \iff z^2 - ax^2 - by^2$ has a non-trivial solution in $\mathbf{Q}_p^3 \iff b$ is a norm from $\mathbf{Q}_p(\sqrt{a})$, where $N(x + y\sqrt{a}) = x^2 - ay^2$.

Theorem 2.1 (Hilbert reciprocity law). For all $a, b \in \mathbf{Q}^\times$,

$$\prod_p (a, b)_p = 1,$$

where the product runs over all primes p , including $p = \infty$.

Note that the product is finite: $(a, b)_p = 1$ for all $p \nmid 2ab$ (see Lemma 2.4 below).

Lemma 2.2. Let $a, b, c \in \mathbf{Q}_p^\times$. Then

1. $(1, c)_p = 1$;
2. $(-c, c)_p = 1$;
3. if $(a, c)_p = 1$, then $(ab, c)_p = (a, c)_p(b, c)_p$ (this holds in general, see the Corollary below);
4. $(c, c)_p = (-1, c)_p$.

Proof. 1. Consider $N : \mathbf{Q}_p(\sqrt{c}) \rightarrow \mathbf{Q}_p$. Then $N(1) = 1$.

2. If $c \in \mathbf{Q}_p^{\times 2}$, then $\mathbf{Q}_p(\sqrt{c}) = \mathbf{Q}_p$, the norm is the identity and $N(-c) = -c$. If $c \notin \mathbf{Q}_p^{\times 2}$, then $N(\sqrt{c}) = -c$. In both cases $-c$ is a norm from $\mathbf{Q}_p(\sqrt{c})$.

3. If $a = N(x)$ and $b = N(y)$, then $ab = N(x)N(y) = N(xy)$, since the norm is multiplicative by definition.

Moreover, if $a = N(x)$, then $a^{-1} = N(x^{-1})$; if in addition $ab = N(z)$, then

$$b = a^{-1}(ab) = N(x^{-1})N(z) = N(x^{-1}z).$$

So, if $(a, c)_p = 1$ (i.e. $a = N(x)$ is a norm from $\mathbf{Q}_p(\sqrt{c})$), then $(b, c)_p = 1 \iff (ab, c)_p = 1$.

4. By (2) and (3),

$$(c, c)_p = (-c, c)_p(c, c)_p = (-c^2, c)_p = (-1, c)_p,$$

because in general $(ad^2, b)_p = (a, b)_p$. □

Remark 2.3. Since $(ad^2, b)_p = (a, b)_p$, the Hilbert symbol is a well-defined map

$$(\cdot, \cdot)_p : (\mathbf{Q}_p^\times / \mathbf{Q}_p^{\times 2}) \times (\mathbf{Q}_p^\times / \mathbf{Q}_p^{\times 2}) \rightarrow \mu_2.$$

Lemma 2.4. Let $p > 2$. Then $(u, v)_p = 1$ for all $u, v \in \mathbf{Z}_p^\times$. (Recall that $\mathbf{Z}_p$ is local and $\mathbf{Z}_p^\times = \mathbf{Z}_p \setminus (p)\cdot$)

Proof. Let $\bar{u} := u \bmod p$ and $\bar{v} := v \bmod p$; they are non-zero elements of $\mathbf{F}_p$ (units of the local ring). We first show that $1 - \bar{u}x^2 - \bar{v}y^2$ has a solution in $\mathbf{F}_p$. Since $\mathbf{F}_p^\times$ is cyclic of order $p-1$,

$$\#\{x^2 : x \in \mathbf{F}_p\} = \frac{p+1}{2}.$$

Similarly, if $S_{\bar{u}} := \{\bar{u}x^2 : x \in \mathbf{F}_p\}$ and $S_{\bar{v}} := \{1 - \bar{v}y^2 : y \in \mathbf{F}_p\}$, then $\#S_{\bar{u}} = \#S_{\bar{v}} = \frac{p+1}{2}$. Thus, $S_{\bar{u}} \cap S_{\bar{v}} \neq \emptyset$ by the

We conclude by Hensel's lemma: □

Lemma 2.5 (Hensel). Let $f \in \mathbf{Z}_p[X]$ and $a_0 \in \mathbf{Z}_p$ with $f(a_0) \equiv 0 \pmod{p}$ and $f'(a_0) \not\equiv 0 \pmod{p}$. Then there exists $a \in \mathbf{Z}_p$ with $f(a) = 0$ and $a \equiv a_0 \pmod{p}$.

This allows us to lift a solution mod p to a solution in $\mathbf{Z}_p$ (to apply it in the proof above, we have to fix one variable).

Proposition 2.6. Let $p > 2$ and $a, b \in \mathbf{Q}_p^\times$, with $a = p^\alpha u$, $b = p^\beta v$ and $u, v \in \mathbf{Z}_p^\times$. Then

$$(a, b)_p = (-1)^{\alpha\beta\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right)^{-\beta} \left(\frac{\bar{v}}{p}\right)^{\alpha}.$$

Proof. Since $(a, b)_p$ depends only on $a, b \bmod \mathbf{Q}_p^{\times 2}$, we may assume $\alpha, \beta \in \{0, 1\}$.

- $\alpha = \beta = 0$: this is Lemma 2.4.
- $\alpha = 1, \beta = 0$: we need $(pu, v)_p = \left(\frac{\bar{v}}{p}\right)$. By Lemma 2.4 we have $(u^{-1}, v)_p = 1$, hence by Lemma 2.2(3)

$$(pu, v)_p = (pu, v)_p \underbrace{(u^{-1}, v)_p}_{=1} = (p, v)_p.$$

If $v \in \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = (p, 1)_p = 1 = \left(\frac{\bar{v}}{p}\right)$ (note that $\bar{v}$ is then also a square mod p). If $v \notin \mathbf{Q}_p^{\times 2}$, then $(p, v)_p = -1$, because $z^2 - px^2 - vy^2$ has no non-trivial solution mod p , hence none in $\mathbf{Q}_p$ by homogeneity; also $\left(\frac{\bar{v}}{p}\right) = -1$ (by Hensel, since $v \notin \mathbf{Q}_p^{\times 2}$).

- $\alpha = \beta = 1$: we need $(pu, pv)_p = (-1)^{\frac{p-1}{2}} \left(\frac{\bar{u}}{p}\right) \left(\frac{\bar{v}}{p}\right)$. By Lemma 2.2(2) we have $(-pv, pv)_p = 1$, so by Lemma 2.2(3)

$$(pu, pv)_p = (pu, pv)_p \underbrace{(-pv, pv)_p}_{=1} = (-p^2 uv, pv)_p = (-uv, pv)_p = (pv, -uv)_p,$$

and now we are in the previous case, which gives

$$(pv, -uv)_p = \left(\frac{-uv}{p}\right) = \left(\frac{-1}{p}\right) \left(\frac{\bar{u}}{p}\right) \left(\frac{\bar{v}}{p}\right),$$

where $\left(\frac{-1}{p}\right) = (-1)^{\frac{p-1}{2}}$, and we used that the Legendre symbol is multiplicative. $\square$

Corollary 2.7. Let $p > 2$. Then $(a, c)_p(b, c)_p = (ab, c)_p$ for all $a, b, c \in \mathbf{Q}_p^\times$. In particular, $(\cdot, \cdot)_p$ is bi-multiplicative.

Proof. Write $a = p^\alpha u$, $b = p^\beta v$ and $c = p^\gamma w$ with $u, v, w \in \mathbf{Z}_p^\times$. By Proposition 2.6,

$$\begin{aligned} (a, c)_p(b, c)_p &= (-1)^{\frac{p-1}{2}\alpha\gamma} \left(\frac{\bar{u}}{p}\right)^\gamma \left(\frac{\bar{w}}{p}\right)^\alpha \cdot (-1)^{\frac{p-1}{2}\beta\gamma} \left(\frac{\bar{v}}{p}\right)^\gamma \left(\frac{\bar{w}}{p}\right)^\beta \\ &= (-1)^{\frac{p-1}{2}\gamma(\alpha+\beta)} \left(\frac{\bar{uv}}{p}\right)^\gamma \left(\frac{\bar{w}}{p}\right)^{\alpha+\beta} = (ab, c)_p. \end{aligned} \quad \square$$

Case $p = 2$. If $a = 2^\alpha u$ and $b = 2^\beta v$ with $u, v \in \mathbf{Z}_2^\times$, define

$$\varepsilon(u) := \frac{u-1}{2} \bmod 2, \quad \omega(u) := \frac{u^2-1}{8} \bmod 2.$$

Then

$$(a, b)_2 = (-1)^{\varepsilon(u)\varepsilon(v) + \alpha\omega(v) + \beta\omega(u)}.$$

(Proof not given.) In particular, $(\cdot, \cdot)_p$ is bi-multiplicative for every p .

Proof of Theorem 2.1. Using bi-multiplicativity, we reduce to the case where each of a, b is either -1 or a prime.

- Case $a = -1$: homework.
- Case $a = b = q$ prime: $(q, q)_p = (-1, q)_p$ by Lemma 2.2(4), so we are reduced to the previous case.
- Case $a = 2$, $b > 2$ prime: $(2, b)_p = 1$ for $p \notin \{2, b\}$ (also for $p = \infty$, as $2, b > 0$). Moreover, by the $p = 2$ formula,

$$(2, b)_2 = (-1)^{\omega(b)},$$

and by Proposition 2.6

$$(2, b)_b = \left(\frac{2}{b}\right) = (-1)^{\frac{b^2-1}{8}}.$$

The product of these two factors is 1.

- Case $a \neq b$ odd primes: $(a, b)_p = 1$ for $p \notin \{2, a, b\}$ (also for $p = \infty$), and

$$(a, b)_p = \begin{cases} (-1)^{\frac{a-1}{2} \frac{b-1}{2}} & p = 2, \\ \left(\frac{a}{b}\right) & p = b, \\ \left(\frac{b}{a}\right) & p = a. \end{cases}$$

Hence the product is 1 by quadratic reciprocity. $\square$

3 Cohomology of groups

Already for the local part of theory we'll require some input from homological algebra, mainly the description of the first cohomology group $H^1(G, A)$ for G some Galois group, and its interpretation in terms of crossed homomorphisms. A concise reference for this material can be found in the notes [2]. We recall some results here for the reader's convenience:

Theorem 3.1 (Cohomology of cyclic groups). Let G be a finite cyclic group of order n , generated by σ . Then,

$$\begin{cases} H^0(G, A) = A^G, \\ H^1(G, A) = A/(1 - \sigma)A, \\ H^2(G, A) = A^G/NA, \\ H^i(G, A) \cong H^{i+2}(G, A) \text{ for all } i \geq 1, \end{cases}$$

This comes out of the “periodic” resolution of $\mathbf{Z}$ as a trivial G -module:

$$\dots \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\sigma-1} \mathbf{Z}[G] \xrightarrow{N} \mathbf{Z}[G] \xrightarrow{\varepsilon} \mathbf{Z} \rightarrow 0,$$

where $N = 1 + \sigma + \dots + \sigma^{n-1}$ is the norm map and ε sends σ to 1. A similar computation holds for the infinite cyclic group $G = \mathbf{Z}$:

$$H^0(\mathbf{Z}, A) = A^{\mathbf{Z}}, \quad H^1(\mathbf{Z}, A) = A/(1 - \sigma)A, \quad H^i(\mathbf{Z}, A) = 0 \text{ for all } i \geq 2.$$

For a subgroup $H \subset G$, we denote by

$$M_H^G(A) := \text{Hom}_H(\mathbf{Z}[G], A)$$

the G -module with action

$$\sigma(\phi)(g) = \phi(\sigma^{-1}g), \quad \sigma \in G, g \in G, \phi \in M_H^G(A).$$

Then, *Shapiro's lemma* tells us that there is a canonical isomorphism

$$H^i(G, M_H^G(A)) \cong H^i(H, A). \quad (3.1)$$

References

- [1] James S. Milne, *Class field theory*, <https://www.jmilne.org/math/CourseNotes/CFT.pdf>.
- [2] Tamás Szamuely, *Notes on homological algebra*, <https://pagine.dm.unipi.it/tamas/ist-alg2new.pdf>.